Aatka

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Education

NED University of Engineering and Technology — Karachi, Pakistan
B.E. Computer & Information Systems Engineering – CGPA: 3.61 — *Oct 2023 – Jun 2027*

BAMM PECHS Govt Girls College — Karachi, Pakistan
Pre-Engineering – 83.8% — *Jun 2021 – Jun 2023*

Experience

Machine Learning Intern — Jul 2026 – Aug 2026
FlyRank AI — *Remote*

- Built a **leakage-safe ML pipeline** to prioritize web pages for content refresh using search and engagement signals.
- Engineered temporal features with **DuckDB** and Python; benchmarked Logistic Regression, Decision Tree, and Random Forest using time-aware validation and ranking metrics.

Co-Head, Coding & Innovation — Apr 2025 – Present
CIS Community, NEDUET — *Karachi, Pakistan*

- Designed and orchestrated **"The Logic Forge,"** a 3-week competitive DSA showdown, and architected **"The CIS Coders Arena,"** a 24-hour live programming event with 3 rounds and 11 distinct challenges.
- Served as **Technical Judge** for the PulseCode Ramadan Hackathon, auditing codebase readability and architecture for AI/ML submissions; recruited and onboarded engineering students to scale the technical team.

Data Analyst Intern — Dec 2025 – Jan 2026
Pakistan Airports Authority — *Karachi, Pakistan*

- Architected an interactive **Power BI Business Intelligence** dashboard tracking 8 airport charge heads, automating visual reporting across a **PKR revenue stream** (local + foreign).
- Delivered granular analyses ranking top 15 airline operators by revenue, isolating concentration risks and seasonal patterns; cross-validated complex data models from raw aviation billing transactions.

Projects

ResearchPilot AI: Secure Multi-Agent Research System | *Python, Google ADK, Gemini, MCP, FAISS, Streamlit*

- Built a capability-separated multi-agent research pipeline with independent research, indexing, and reporting agents; isolated vector-store persistence behind **FastMCP**.
- Implemented prompt-injection sanitization, citation validation, structured Pydantic outputs, and a Streamlit interface for grounded research reports and presentations.

Sentinel Vision: Multi-Agent AI Governance Pipeline | *React, Gemini 1.5 Flash, GCP, NIST*

- Engineered a real-time AI governance layer orchestrating a sequential **3-agent pipeline** (Sentry, Auditor, Coordinator) to audit prompts for PII leaks and prompt injections; deployed globally via **Google Cloud Run**.
- Earned **Silver Tier recognition** in the national Google AI Seekho Phase 1 challenge, outperforming thousands of builders across Pakistan.

Trail Vision: Semantic Segmentation | *TensorFlow/Keras, U-Net, YOLOv8, MobileNetV2*

- Curated a custom dataset of **613 images / 6,411 polygon instances** across 6 terrain classes; trained 3 architectures (custom U-Net, YOLOv8, ResNet50/MobileNetV2 transfer learning).
- Achieved **91.13% validation accuracy** via stratified splits, 3× augmentation pipeline, and 3× class-weighted Dice loss.

MEDICO: Hospital Management + AI Health Module | *Flask, MySQL, Streamlit, GROQ/LLaMA*

- Architected a role-based full-stack web app with secure session authentication for admins, doctors, and patients; built a decoupled **AI Fitness Advisor** generating real-time personalized dietary and wellness plans via LLaMA.

Achievements

Meta Hacker Cup 2025 (Global): Advanced to Round 2 (Human Track); scored 79/105; ranked **885th out of 13,779 participants** – top **6.5%** worldwide.

Google AI Seekho Phase 1: Silver Tier Swag for Sentinel Vision; top builder among thousands of competitors across Pakistan.

Technical Skills

Languages & Data: Python, SQL, NumPy, Pandas, DuckDB, PostgreSQL
ML: Scikit-Learn, PyTorch, TensorFlow/Keras, Feature Engineering, Model Evaluation
GenAI: LLMs, RAG, LangChain, LlamaIndex, AI Agents, MCP, FAISS, ChromaDB, Embeddings, Reranking
MLOps: MLflow, DVC, Docker, GitHub Actions, Model Monitoring
Engineering: FastAPI, Flask, Streamlit, React, Git/GitHub, Google Cloud Run, Power BI

Certifications

DataCamp: Machine Learning Engineer Career Track (12 courses) | **Cohere Labs:** ML Summer School 2026 | **Kaggle & Google:** 5-Day AI Agents Intensive | **NCEAC / PakAngels HEC:** Generative AI Application Developer | **IBM:** RAG & Agentic AI (Coursera)